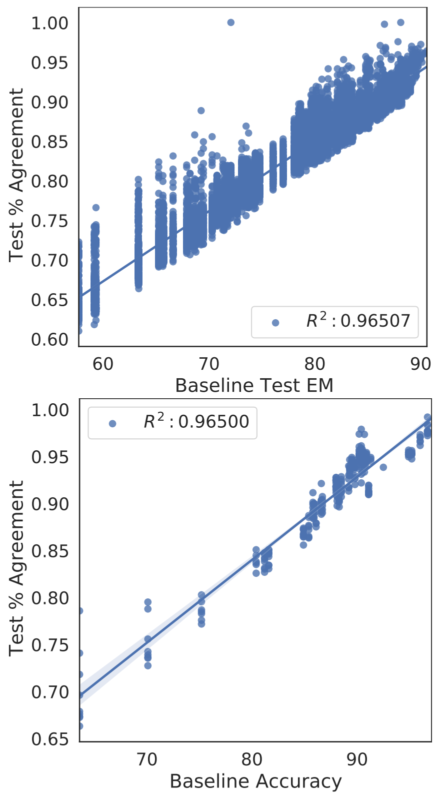

Figure 11: SQuAD 2.0 (top) and GLUE (bottom) % agreement of new model vs. the accuracy of the baseline in the comparison (assuming improvement in the new model).

ratio, the latter being:

$$\Phi = \frac{1 - \text{exp\_overlap} + \text{exp\_acc\_diff}}{1 - \text{exp\_overlap} - \text{exp\_acc\_diff}} \tag{3}$$

This is all that is necessary for McNemar's test and thus we can then simply solve for the minimum expect treatment effect for the given sample size of the dataset and a power of 80%. Note that for QQP we use the normal approximation rather than exact unconditional test as the large sample size makes the exact test intractable. See Duffy (1984).

We fit such a regression to GLUE tasks and achieve an R^2 of 0.97. Repeating this for SQuAD 2.0, we get an R^2 of 0.94, with fit shown in Table 6. See Figure 11 for a plot indicating the level of agreement plotted against baseline accuracy. See also additional model comparisons for overlap in Appendix I.

E.2.2 Predicting Effect Size

A similar regression can be run to predict the expected effect size given the baseline accuracy: how much do models typically improve given the current SOTA. To fit an OLS regression predicting this value, we gather data from GLUE papers regarding the accuracy tasks and manually label 119 comparisons and 57 claims of improvement (as denoted within a work by bolding of a new model's number and a claim of SOTA in the main text) across 14 papers (selected as being at or above the BERT score on the GLUE leaderboard with an accompanying publication). We fit the regression:

$$\widehat{\Delta}_i = \beta_0 + \beta_1 \text{baseline}_i + \hat{\beta}_2 \text{task}_i, \tag{4}$$

to see how predictable the expected effect size is, where $\widehat{\Delta}_i$ is the predicted effect size, baseline$_i$ is the baseline model's accuracy, and task$_i$ is a categorical variable (in the regression this ends up being a set of dummy variables for each category so we denote $\hat{\beta}$ to emphasize this). Note that for SQuAD 2.0, we use a separate regression without the task variable since it is a single-task leaderboard.

We achieve an $R^2 = 0.69$ which is not a perfect fit, but still provides a prior on likely effect size. Similarly, we achieve an $R^2 = .67$ when fitting a regression to SOTA improvements on the SQuAD 2.0 leaderboard (selected as being a significant improvement in time-ordered submissions).

See Table 7 and Table 8 for regression coefficients and model fits. Figure 13 shows the per-task distribution of effect sizes against baseline accuracies in GLUE papers for SOTA improvements. Figure 12 shows the effect size distribution as a histogram.

E.2.3 Caveats for Regression-based Approach

Fitting a regression to predict overlap between a baseline and a new model has a good linear fit. However, this may not be the case for every dataset. Additionally, predicting effect sizes via a linear fit is not a perfect prior. The measurements of power in this case are meant to simulate estimating power before running evaluation on a test set, as running power analysis using only the observed effect may lead to the issues of post-hoc power estimation.

E.3 No Prior Approach (Lachenbruch, 1992)

What do you do if there is no prior data available (as in a new task) and so you cannot make assumptions about discordant pairs or odds ratio? Lachenbruch (1992) discusses this exact problem in the context of clinical trials, and proposes an alternative method based on the work of (Connett et al., 1987) which allows you to make

	Dependent variable:
	effect.size
Previous.Best	-0.264^{***}
	(0.032)
TaskMNLI-mm	0.150
	(0.621)
TaskMRPC	0.023
	(0.622)
TaskQNLI	2.139^{***}
	(0.639)
TaskQQP	-0.195
	(0.719)
TaskRTE	1.018
	(0.628)
TaskSST-2	1.536^{**}
	(0.686)
TaskWNLI	-0.520
	(0.789)
Constant	24.342^{***}
	(2.837)
Observations	61
R^2	0.690
Adjusted R^2	0.642
Residual Std. Error	1.309 (df = 52)
F Statistic	14.455^{***} (df = 8; 52)
Note:	*p<0.1; **p<0.05; ***p<0.01

Table 7: OLS regression for predicting effect size for GLUE tasks.

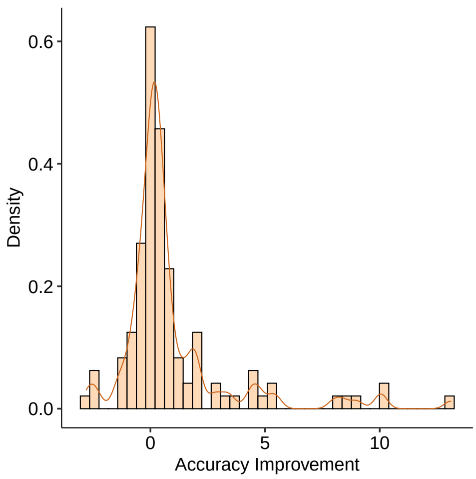

Figure 12: The reported difference from the best performing new model to the best performing baseline in accuracy across all accuracy datasets in the GLUE Benchmark. Note: unlike Table 10, we do not limit these to claims of improvement, but only to papers which introduce a new model and compare against some baseline. Mean: +0.959 Std.Err.: 0.23

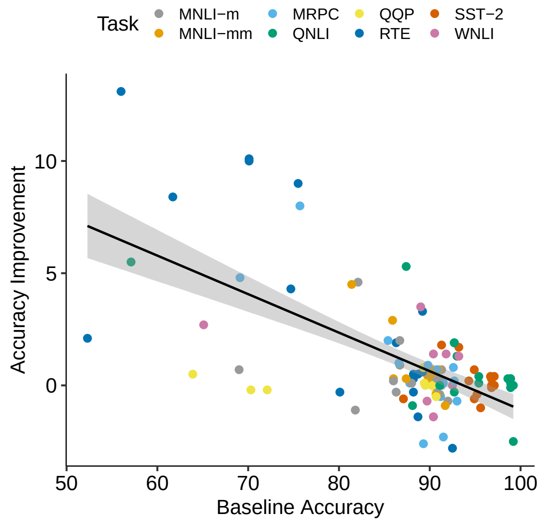

Figure 13: The effect size given the baseline model accuracy observed across GLUE tasks. As the baseline model moves toward the range of current GLUE submissions, reported model gains decrease toward 0. Fitting a regression yields an $R^2 = 0.69$.

assumptions about potential marginal probabilities, providing a midpoint value, as well as an upper and lower bound. We use an implementation of this from: `https://rdrr.io/rforge/biostatUZH/man/sampleSizeMcNemar.html` and solve for the expected accuracy minimum given a fixed dataset sample size and baseline accuracy for each of the lower bound, midpoint, and upper bound. In practice, we find the Lachenbruch (1992) prior to be very close to the values we obtain from the above regression (see Table 9). Importantly this method requires no assumptions and is meant to give an idea for whether it is worth pursuing a study for the given size of the test set.

E.4 Extended Results

Table 9 contains additional MDE estimates using a two-sample proportion test as in Appendix E.5, the Lachenbruch (1992) methodology. We also provide the standard errors and n for each average effect size, the OLS regression predicting the next effect size for a new SOTA $\widehat{\Delta}$, and the current difference from SOTA and next on the leaderboard. We note that MDE calculations are roughly similar except for the upper and lower bounds provided in the Lachenbruch (1992) calculation. We also note that predicted SOTA results are far lower than past averages since the average includes early large results like those of Devlin et al. (2019). We can see that in some cases the predicted effect size

Dep. Variable:	y	**R-squared:**	0.672
Model:	OLS	**Adj. R-squared:**	0.644
Method:	Least Squares	**F-statistic:**	24.55
Date:	Tue, 26 May 2020	**Prob (F-statistic):**	0.000334
Time:	06:05:23	**Log-Likelihood:**	45.711
No. Observations:	14	**AIC:**	-87.42
Df Residuals:	12	**BIC:**	-86.14
Df Model:	1		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	0.1331	0.023	5.910	0.000	0.084	0.182
x1	-0.1408	0.028	-4.955	0.000	-0.203	-0.079

Omnibus:	19.911	**Durbin-Watson:**	2.643
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	18.487
Skew:	1.995	**Prob(JB):**	9.68e-05
Kurtosis:	6.971	**Cond. No.**	17.3

Table 8: OLS Regression Results for predicting effect size from baseline accuracy for SQuAD 2.0 improvements.

is even smaller than the lowest bound MDE and we may wish to consider the usefulness of further comparisons on individual datasets in such cases.

E.5 Calculating Power or Sample Size with Binomial Test

If we assume that samples are *unpaired* – the new model and baseline evaluation samples are drawn from the same data distribution but aren't necessarily the same samples – we can use a binomial test for significance.

In this case, we assume that we have two models and each draw brings a 1 if the model is correct or 0 if incorrect. We would like to use the two-sample proportion test, and have two binomial distributions with p_1 and p_2 as the mean probabilities. Our null hypothesis is $H_0 : p_1 = p_2$. We have an alternative hypothesis (two sided) is $H_1 : p_1 \neq p_2$. Note, in R we can use the function **power.prop.test()** to calculate power, the MDE, or the sample size of the tests. See also a tutorial here: `https://imai.fas.harvard.edu/teaching/files/Handout9.pdf`.

F Additional Metrics

In this appendix, we provide guidance on how we might apply power analysis to metrics beyond what is covered in the main paper.

Recall, Precision, F1, Matthew's correlation: While accuracy is the most commonly used metric in the GLUE benchmark, other tasks make use of other metrics such as F1 and Matthew's correlation. F1 is particularly relevant in cases of binary classification where there is strong class imbalance, such that even the baseline of predicting the most common class will achieve high accuracy.

If we have good prior information, we can use an approach akin to that recommended for accuracy, but replacing McNemar's test with a randomization test (as used for machine translation, see §4 in main paper). In particular, given an evaluation on paired data (as is the case for all benchmark datasets), one can test for a significant difference between models in terms of F1 (or any other metric) using a randomization test. That is, on each iteration, we randomize the assignment of which model each prediction came from for every instance with probability 0.5, and compute the resulting overall difference in F1. Repeating this thousands of times gives us the null distribution, and we can then check to see whether the observe difference in F1 is in the tails of this distribution, which can thereby be converted into a p-value (see Dror et al. (2018) for more details).

Because F1 (and related metrics) cannot be represented as a simple sum over individual instances, in order to completely specify a hypothetical data generating process, we need to assume values for all cells in the confusion matrix, per class. That is for each class we would need to assume values for the cells as shown in Table 11, where the relevant distribution of predictions are for the instances with the corresponding label, and the values for